

PS3 makes this prof see stars

With the computing power of 25 desktops, the PlayStation 3 now doubles up as a scientific research tool

How would you use a PlayStation 3? Sounds like a silly question because everyone knows Sony's PS3 is for gaming. But a physics professor in the US has other uses for it.

Gaurav Khanna of the University of Massachusetts is using the PS3 not to play Tekken 5, but to unravel complex problems in astrophysics — “simulations to understand what gravity waves would look like if a small star gets captured by a black hole,” Khanna told DNA.

Why the PS3? Khanna needed phenomenal computing power to conduct his simulations. One option was to connect desktop computers in parallel and use their combined processing power. “I would've needed 200 desktop PCs for the purpose. The cost and space issues along with power consumption made that totally impractical,” he says. The other option was to use a supercomputer. “Time constraint was the problem with a mainframe computer. You have to pay for specific time slots, and that didn't suit me either.”

It was then that Khanna decided to try out the PS3, which has 25 times the computing power of a desktop PC thanks to its processor dubbed ‘a supercomputer on a chip’.

Developed jointly by IBM, Sony and Toshiba, it has eight processing units capable of handling separate computing tasks simultaneously.

Khanna put together eight PS3s in a grid, and presto, he had something close to a supercomputer at his command, at a small fraction of its cost. “Its computational power makes it possible to do things that were difficult earlier,” Khanna says of the PS3.

Khanna ran the Linux operating system on the PS3. “Given the open PS3 platform and the legendary performance of the Cell processor that powers a PS3, it was natural to

consider it for scientific research,” he says.

So while his daughter enjoys Tekken 5 on the console (Prof Khanna has one at home for gaming), he puts the same processor to task to understand the mysteries of the universe.